Enhancing Scene Generalization for Open-Vocabulary Remote Sensing Segmentation via Semantic–Structural Collaboration

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***Abstract*—Open-vocabulary semantic segmentation (OVSS) of remote sensing faces severe performance degradation when encountering unseen scene distributions caused by geographic, sensor, and resolution variations. Existing vision–language approaches provide strong semantic priors but lack scene-invariant structural representations required for dense prediction. In this work, we focus on open-vocabulary scene generalization semantic segmentation (OVSGSS), achieving dense inference guided by joint modeling of semantic alignment and structural consistency. To this end, we construct USGMS-100K, a large-scale multisensor dataset for self-supervised pretraining, and develop a structure-aware remote sensing image encoder (RSIE) via masked reconstruction to learn scene-invariant representations. Building upon this encoder, we propose a semantic–structural collaborative framework (namely RS-OVSGSeg) that integrates language-derived semantic priors with structural priors through a semantic–structural cost map enhancement (SSCME) module and a dual-prior guided decoder (DPGD). Extensive cross-scene evaluations on five public datasets demonstrate that proposed method achieves state-of-the-art performance in open-vocabulary cross-scene segmentation, while maintaining a favorable balance between accuracy and computational efficiency. The results highlight the importance of explicitly modeling structural invariance for robust open-vocabulary scene generalization in remote sensing. The USGMS-100K dataset, RSIE weight and code are publicly available at https://github.com/HuangWBill/RS-OVSGSeg.**

***Index Terms*—Mask self-supervision, open-vocabulary semantic segmentation (OVSS), remote sensing images, scene generalization, semantic–structural collaboration.**

I. Introduction

REMOTE sensing image semantic segmentation (RSISS) is one of the fundamental tasks in Earth observation, playing a crucial role in applications such as land cover classification [1], agricultural monitoring [2], and disaster assessment [3]. Unlike image-level classification or object detection, semantic segmentation requires dense, pixel-wise, and spatially consistent predictions, making it particularly sensitive to variations in scene appearance. With the continuous expansion of remote sensing platforms and data sources, achieving stable and reliable semantic segmentation across complex and diverse scenarios remains a long-standing and critical challenge.

Despite significant progress in supervised semantic segmentation in recent years, most existing methods heavily rely on large-scale pixel-level annotations and are typically trained under a closed-set assumption [4], [5]. In real-world remote sensing applications, obtaining exhaustive pixel-wise annotations across different geographic regions, sensors, and spatial resolutions is extremely costly and labor-intensive. Moreover, segmentation models trained on specific datasets often suffer from severe performance degradation when deployed to unseen scenes [6], [7], limiting their practical generalization capability.

These limitations have motivated researchers to explore more flexible segmentation paradigms that can go beyond predefined label sets and training scenarios. In this context, open-vocabulary semantic segmentation (OVSS) has emerged as a promising research direction, aiming to recognize and segment previously unseen classes through semantic guidance rather than explicit pixel-level supervision [8]. By using vision–language pretrained models such as contrastive language-image pretraining model (CLIP) [9], OVSS methods align visual representations with text descriptions and have demonstrated strong zero-shot generalization ability in natural images.

However, directly transferring these methods to remote sensing image remains highly challenging. Compared to natural-scene images, remote sensing scenes exhibit much stronger cross-scene heterogeneity due to variations in geographic regions, imaging platforms, and spatial resolutions. Such heterogeneity makes models more prone to performance degradation when encountering unseen scenarios. A key reason is that existing CLIP-based OVSS frameworks are primarily optimized for high-level semantic abstraction, emphasizing global semantic representations while lacking sufficient modeling of fine-grained spatial structures and mul-

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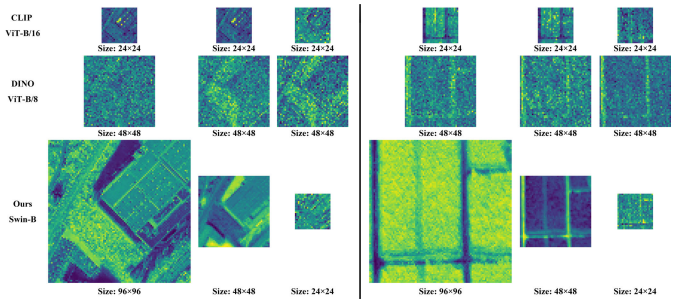


Fig. 1. Visualization of multilevel features extracted by different pretrained visual encoders for a 384 × 384 remote sensing image. CLIP (ViT-B/16) produces uniformly low-resolution features (24 × 24) across layers, while GSNet (ViT-B/8) increases spatial resolution at the cost of significantly higher computation. In contrast, the proposed Swin-based RSIE provides hierarchical multiscale features (96 × 96, 48 × 48, 24 × 24), which are more suitable for dense prediction and scene generalization in remote sensing.

tiscale layouts [10]. In remote sensing images, many semantic classes are inherently scene-level concepts whose discrimination relies heavily on multiscale structural relationships and spatial organization. Consequently, relying solely on global semantic alignment is insufficient for robust segmentation.

To address these issues, recent studies [11], [12] have attempted to incorporate self-supervised pretraining techniques such as self-Distillation with No Labels (DINO) [13] into OVSS frameworks to enhance visual representation robustness. Although these methods have improved feature quality to some extent, they typically employ vision transformer (ViT)-based encoders with fixed patch sizes and uniform feature resolutions (as showed in Fig. 1). Such architectures still face inherent limitations in modeling hierarchical spatial structures and low-level geometric details. Increasing patch resolution can improve spatial granularity but incurs substantial computational and memory overhead. Moreover, the unified multiscale ViT features require complex upsampling operations during the decoding stage, which still make it difficult to fully recover the structural information critical for RSISS.

Motivated by the above analysis, we revisit the problem of OVSS in remote sensing from a scene generalization perspective and propose a novel framework for open-vocabulary scene generalization semantic segmentation (OVSGSS) in remote sensing images. In this work, scene generalization specifically refers to the ability of a model trained on a source remote sensing dataset to generalize to unseen datasets with different geographic regions, imaging conditions, spatial resolutions, and land-cover compositions. In contrast, conventional remote sensing domain generalization tasks may additionally involve distribution shifts caused by cross-modality settings (e.g., RGB-to-NIR) or different sensors. Furthermore, under the open-vocabulary setting, the target datasets may contain semantic classes that are absent from the training data. Therefore, OVSGSS requires not only scene generalization across unseen datasets but also semantic generalization to unseen classes. Under this setting, we investigate OVSGSS from a semantic–structural collaborative modeling perspective to improve the robustness of open-vocabulary segmentation under scene variations. The main contributions of this article are summarized as follows.

1) We propose a novel semantic–structural collaborative framework, named RS-OVSGSeg. The proposed framework explicitly models the collaboration between semantic priors and scene-invariant structural representations, thereby improving robustness under scene variations and enhancing generalization to unseen classes.
2) To learn scene-invariant structural representations, we construct a large-scale unsupervised multisensor dataset, namely USGMS-100K. Based on this dataset, we pretrain a Remote Sensing Image Encoder (RSIE) using masked self-supervised learning, enabling the extraction of hierarchical, multiscale, and structure-aware visual representations that are highly transferable across unseen scenes.
3) We design a semantic–structural cost map enhancer (SSCME) and a dual-prior guided decoder (DPGD) to perform semantic–structural interaction and progressive refinement. SSCME injects scene-invariant structural priors into semantic cost maps to alleviate semantic ambiguity caused by scene variations, while DPGD progressively integrates semantic and structural guidance during decoding to preserve spatial continuity, boundary integrity, and fine-grained class discrimination.
4) We conduct extensive experiments on multiple representative datasets. The results demonstrate that the proposed method achieves more robust and superior performance in unseen class segmentation and challenging cross-scene settings compared to existing methods.

The remainder of this article is organized as follows. Section II reviews related work on OVSS in remote sensing, and scene generalization. Section III presents the proposed RS-OVSGSeg framework for remote sensing in detail. Section IV describes the datasets, experimental settings, and evaluation metrics. Section V analyzes the experimental results and ablation studies. Section VI discusses the effectiveness and limitations of introducing the RSIE. Finally, Section VII concludes this article.

II. Related Work

A. OVSS for Remote Sensing

In recent years, OVSS for remote sensing has attracted increasing attention. Cat-Seg [14] first demonstrated the feasibility of OVSS through vision–language alignment. Subsequently, many studies have explored extending the OVSS paradigm to remote sensing images, existing methods can be broadly categorized into training-free methods and training-based methods.

SegEarth-OV [15] is the most representative training-free OVSS method, which reconstructs high-resolution segmentation outputs by upsampling low-resolution semantic features, thereby recovering spatial details lost in deep representations while preserving semantic consistency. SegEarth-OV3 [16] further investigates the application of Segment Anything Model 3 (SAM 3) [17] to remote sensing OVSS by integrating

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the advantages of both the semantic head and instance head of SAM. Although such methods are appealing due to their zero training cost and strong generalizability, their performance is fixed after deployment and highly sensitive to hyperparameter selection. Moreover, they often incur substantial computational and time costs during the inference process.

In contrast, training-based OVSS methods introduce learnable decoders and aggregation modules that are optimized on specific datasets. These methods generally follow the CatSeg framework and focus on improving OVSS performance by enhancing either text semantic or visual feature representations. To address semantic ambiguity and representation insufficiency arising from single-word class embeddings in remote sensing, TACOSS [18] and TPOV-Seg [19] enrich class representations through synonym expansion and descriptive prompting, respectively. However, these approaches do not explicitly consider the hierarchical relationships among semantic classes. KG-OVRSeg [20] introduces a land-cover concept knowledge graph to explicitly encode semantic hierarchies between classes, effectively alleviating ambiguity in text embeddings. In this work, we adopt the knowledge graph-enhanced class encoder (KGCE) proposed in KG-OVRSeg as our text embedding module to mitigate semantic ambiguity in remote sensing classes, allowing us to focus on visual generalization challenges.

Regarding visual feature representation, recent methods primarily aim to compensate for the limitations of the CLIP visual encoder when applied to remote sensing images. OVRS [21] introduces rotation aggregation and multiscale fusion to address common orientation and scale variations in remote sensing scenes. However, it relies solely on CLIP visual features pretrained on natural images, which lack remote sensing-specific structural priors. GSNet [11] uses pretrained DINO features on the LandDiscover50K dataset as a domain expert to enhance feature fusion and decoding. AerOSeg [22] employs SAM features to guide spatial refinement of relevant features. Nevertheless, these methods largely overlook key factors such as large-scale contextual information and object rotation. RSKT-Seg [12] further integrates rotation aggregation, DINO-based domain experts, and RemoteCLIP visual features for upsampling and fusion, yielding more robust and accurate image representations. Beyond incorporating additional domain-specific encoders, Skysense-O [10] proposes a SwinV2-Huge-based remote sensing vision–language model that unifies visual self-supervised learning and vision–language pretraining, demonstrating strong performance across multiple downstream tasks including OVSS. From a weakly supervised perspective, T2ASeg [23] adopts text-guided adaptive pseudo-label generation to jointly model semantic consistency and spatial structure under noisy supervision.

Despite these advancements, most existing training-based OVSS methods still rely on ViT-style visual encoders with limited multiscale structural modeling capability or introduce multiple specialized encoders that substantially increase model complexity. More importantly, few studies explicitly address scene generalization, i.e., maintaining stable OVSS performance under significant variations in geographic regions, sensors, and spatial resolutions. This gap motivates our work, which focuses on learning robust, structure-aware visual representations through self-supervised pretraining and integrating them with CLIP-based semantic priors to enhance OVSGSS in remote sensing images.

B. Scene Generalization in Remote Sensing

Scene generalization has long been a fundamental challenge in remote sensing image understanding. Due to variations in geographic distribution, sensor types, and acquisition conditions, remote sensing images exhibit significant heterogeneity, which severely limits the generalization capability of data-driven models to unseen scenes. Early studies primarily assumed that target scene data were accessible during training, leading to the prevalent use of domain adaptation techniques. Representative approaches align feature distributions between source and target scenes through feature space alignment [24], GAN-based style transfer [25], or adversarial learning with scene classifiers that encourage scene-invariant representations [26]. However, these methods require access to target scene data during training, which is often unrealistic in real-world remote sensing applications.

Consequently, subsequent research has shifted toward scene generalization, which aims to improve robustness to unseen scenes without accessing target scene data. A common strategy is to simulate scene variations via data augmentation and style randomization, encouraging models to learn more generalizable representations. CCDR [27] introduces texture and style randomization for auxiliary joint supervision. GeoSA-BaSA [7] performs batch-level style mixing in the frequency domain for data augmentation. Beyond data-level augmentation, several studies focus on learning scene-invariant representations at the feature level. FOSMix [28] generates mixed data through frequency-scene style randomization and employs consistency constraints to learn domain-invariant features. Wang et al. [29] design style prototypes to store source scene style statistics and align shallow features of target scenes with these prototypes during inference, preserving prior knowledge. OCDA [30] proposes a two-stage open composite domain adaptation framework, where images from different styles are first translated into a unified appearance space before learning class-specific semantic representations.

Recently, self-supervised learning has emerged as a promising paradigm for improving scene generalization in remote sensing. By utilizing large-scale unlabeled data, self-supervised pretraining enables models to learn more robust and transferable visual representations. Some works explore parameter-efficient adaptation strategies based on self-supervised pretrained models. Scheibenreif et al. [31] propose a continual pretraining framework that trains adapters in a self-supervised manner on target scenes. Earth-Adapter [32] and GeoSA-BaSA [7] apply parameter-efficient fine-tuning to adapt DINOv2 to remote sensing scenes. CrossEarth [33] designs a cross-domain generalization foundation model using masked modeling and style injection to enhance robustness across diverse scenes.

However, most existing scene generalization studies in remote sensing operate under a closed-set assumption, where

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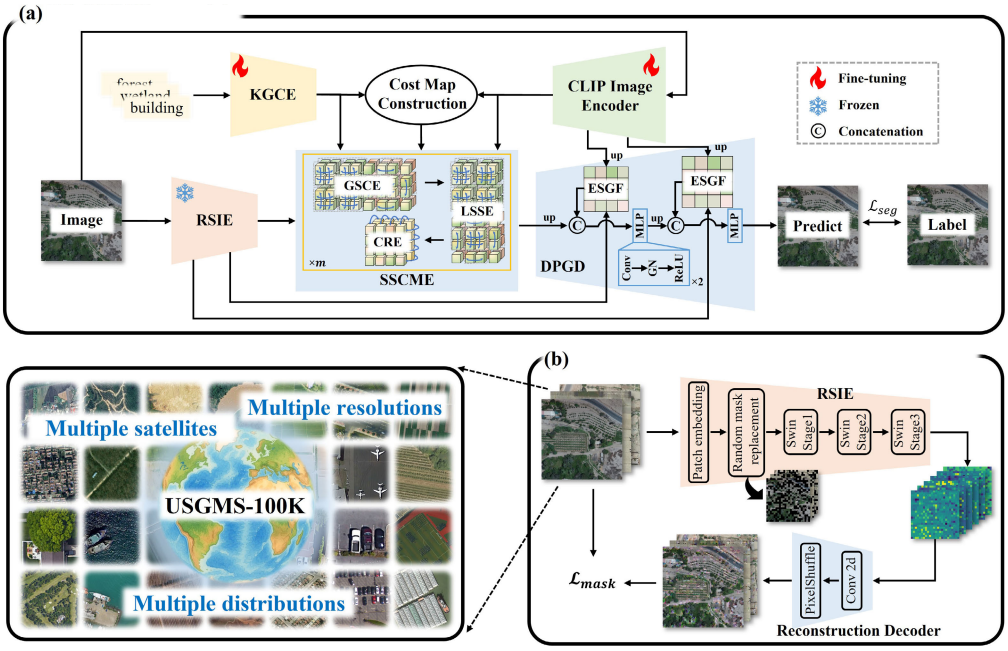


Fig. 2. Overall framework of the proposed method. (a) Proposed RS-OVSGSeg model, (b) masked self-supervised pretraining of RSIE.

the label space remains fixed across domains. In contrast, OVSGSS requires models to handle both unseen classes and unseen scenes simultaneously, posing a more complex challenge. The interaction between semantic transfer and spatial generalization has not been fully explored in current research.

III. Methodology

A. Method Overview

Fig. 2 shows the overall pipeline of the proposed RS-OVSGSeg framework for OVSGSS in remote sensing images. As shown in Fig. 2(a), RS-OVSGSeg explicitly models the collaboration between semantic priors derived from vision–language pretraining and scene-invariant structural representations learned from large-scale remote sensing data. The overall objective is to simultaneously improve semantic generalization to unseen classes and robustness under scene variations.

Given a set of batch input image $I \in \mathbb{R}^{B\times H\times W\times 3}$, RS-OVSGSeg employs two parallel visual encoders, CLIP image encoder and RSIE, to capture different representations. The CLIP image encoder adopts a ViT-B/16 architecture and is fine-tuned using attention tuning [14] during the training process to extract high-level semantic features $F_{clip} = \{F^1_{clip} \in \mathbb{R}^{B\times C_3\times H/16\times W/16}, F^2_{clip} \in \mathbb{R}^{B\times C_3\times H/16\times W/16}, F^3_{clip} \in \mathbb{R}^{B\times C\times H/16\times W/16}\}$, where F^3_{clip} is strongly aligned with text embeddings and serves as the primary source of semantic guidance. RSIE uses a Swin-B architecture pretrained on the proposed USGMS-100K using masked self-supervised learning and remains frozen during training. It generates multiscale structure-aware features $F_{rs} = \{F^1_{rs}, F^2_{rs}, F^3_{rs}\}$.

For any set of class names $T = \{t_1, t_2, \ldots, t_N\}$, we use the KGCE proposed in our previous work [20] to obtain class embeddings $F_{class} \in \mathbb{R}^{B\times N\times P\times C}$, where P is the number of prompt templates. Since KGCE relies on pretrained CLIP text encoders, we do not extend the class names during the training process and use attention fine-tuning to ensure alignment between text and image features. During the inference process, KGCE aggregates structured semantic relationships among classes to generate more discriminative and consistent class representations. Since the effectiveness of KGCE has been thoroughly validated in previous work, we do not elaborate further here.

Based on the semantic similarities between visual and text features, an initial semantic cost map F_{cost} is first constructed. Subsequently, the proposed SSCME injects scene-invariant structural priors into the semantic cost map to alleviate semantic ambiguities caused by scene variations. Finally, the DPGD progressively integrates semantic and structural guidance to generate dense segmentation predictions with enhanced spatial continuity, boundary integrity, and fine-grained class discrimination under open-vocabulary and unseen-scene settings.

The remainder of this section is organized as follows. Section III-B describes the construction process of the proposed USGMS-100K dataset and the self-supervised pretraining strategy of RSIE. The semantic cost map construction, SSCME, and DPGD are detailed in Sections III-C, III-D, and III-E, respectively.

B. USGMS-100K and RSIE Pretraining

1) USGMS-100K: A large-scale unsupervised scene generalization of multisensor dataset: In recent years, various publicly available remote sensing datasets have been released, covering different sensors, spatial resolutions, and geographic regions. To effectively utilize existing data while addressing OVSGSS challenges, we select four representative public datasets as the sources for constructing a unified unsupervised dataset. The selection is guided by three key criteria: 1) diversity of satellite types; 2) variation in spatial resolution;

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TABLE I
COMPOSITION OF THE USGMS-100K DATASET

Dataset	Year	Source	Spatial resolution (m)	Coverage	Selected Images	Generated Patches	License
OpenEarthMap	2023	Satellite and Aerial	0.25-0.5	97 regions across 44 countries on six continents	3500	13629	CC BY-NC-SA 4.0
EvLab-SS	2017	Wordview-2, GeoEye, QurckBird, GF-2, Aerial	0.1-2	China	60	10914	Obtained author email permission
CASID	2023	Google Earth	1	30 different cities/regions in Asia	90	17640	MIT
DIOR	2018	Google Earth	0.5-30	Over 80 countries worldwide	23432	57817	CC BY-NC 4.0

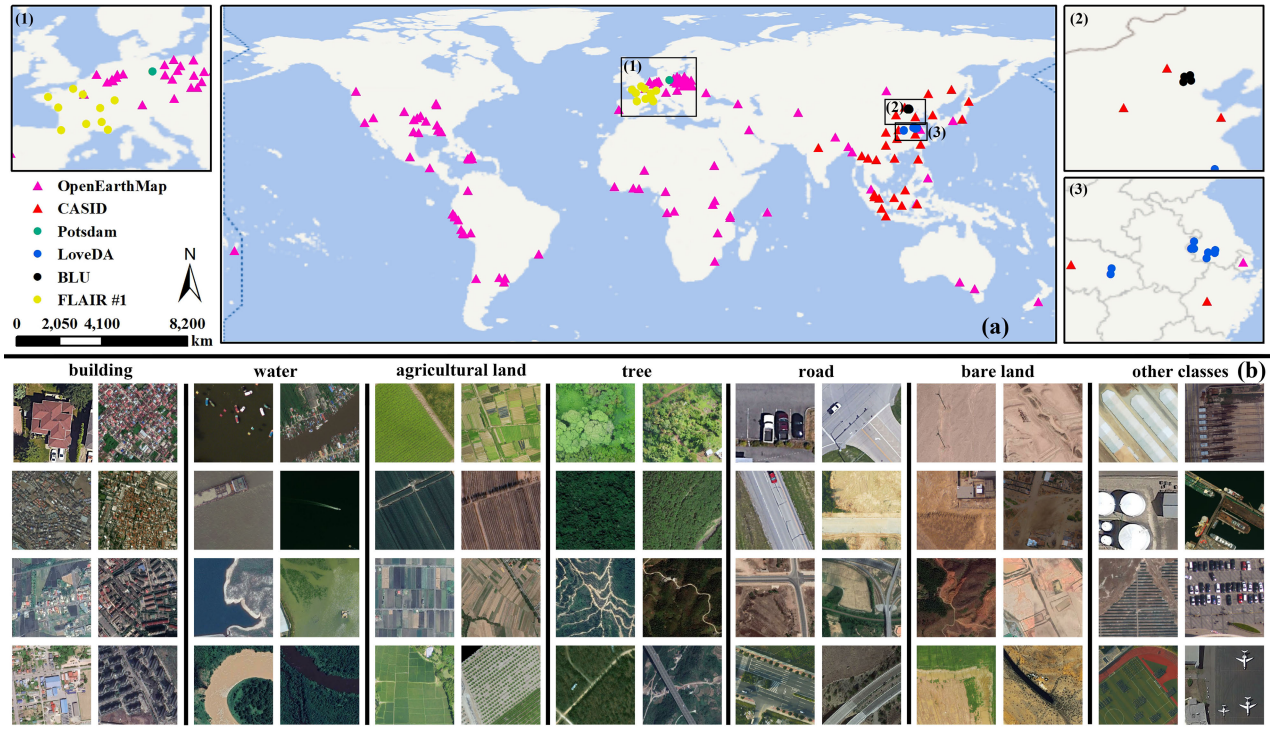


Fig. 3. (a) Geographic distribution of all datasets used in this study. Triangles represent datasets incorporated into USGMS-100K, whereas circles denote datasets used solely for evaluating cross-dataset generalization. The geographic information of EvLab-SS, DIOR, and IRSAMap is not publicly available and is therefore excluded from the map. (b) Representative samples from USGMS-100K, demonstrating substantial diversity in geographic coverage, scene characteristics, spatial resolutions, and semantic classes.

and 3) widespread geographic distribution. Table I and Fig. 3(a) summarizes the characteristics of the selected datasets and the number of images used.

a) OpenEarthMap dataset [34]: This a global high-resolution land cover mapping benchmark, covering 97 regions across 44 countries across six continents. It contains 5000 aerial and satellite images with a total coverage area of approximately 799 km^2. The dataset is characterized by strong geographic diversity and heterogeneous data sources, making it well suited for learning globally generalizable representations.

b) EvLab-SS dataset [35]: It is a high-resolution RSISS dataset designed for real-world engineering scenarios, primarily covering regions in China. These images were collected from multiple platforms, including WorldView-2, GeoEye, QuickBird, GF-2 satellites, and aerial sensors. The dataset has rich semantic classes, diverse imaging conditions, and complex urban and suburban scenes, providing representative real-world variability.

c) CASID dataset [36]: This a large-scale climate-aware satellite image dataset comprising 980 images from 30 regions across Asia, covering more than 24 500 km^2. CASID includes four typical climate zones: temperate monsoon, subtropical monsoon, tropical monsoon, and tropical rainforest, each forming an independent subdomain. The dataset exhibits substantial interdomain differences in texture, spectral characteristics, and object shapes.

d) DIOR dataset [37]: It is a large-scale optical remote sensing object detection benchmark dataset. It contains 23 463 images covering 20 common object classes. The data are primarily sourced from Google Earth and exhibit significant variations in object scale, imaging conditions, season, and weather, making it a challenging and diverse dataset.

For the above datasets, First, representative images are selected from each source dataset. Subsequently, all selected images are partitioned into fixed-size image patches using a nonoverlapping sliding-window strategy with a patch size of 384 $\times$ 384 pixels. As shown in Table I, for OpenEarthMap,

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we use all 3500 images from the training and validation sets, generating 13 629 image patches. For EvLab-SS, all 60 available images are partitioned, resulting in 10 914 patches. For CASID, three images are randomly selected from each of the 30 geographic regions and further cropped into patches, producing 17 640 samples. For DIOR, two diagonal patches are extracted from each image. To balance the dataset size, an additional 5445 upper-right patches are randomly sampled from the training set and 5446 lower-left patches are randomly sampled from the test set, resulting in 57 817 image patches.

After patch generation, all samples are uniformly renamed using six-digit identifiers ranging from 000001 to 100 000. The final USGMS-100K dataset contains exactly 100 000 image patches, each with a fixed resolution of 384 × 384 pixels. As illustrated in Fig. 3(b), the dataset includes both satellite and aerial images, diverse land cover types, multiple spatial resolutions, and rich geographic variability. These characteristics enable USGMS-100K to serve as a large-scale pretraining corpus for learning remote sensing invariant embeddings and improving scene generalization capability under open-vocabulary settings.

2) Masked Self-Supervised Pretraining of RSIE: Based on the constructed USGMS-100K dataset, we pretrain a RSIE using a masked self-supervised learning strategy. By reconstructing masked image regions, the encoder is encouraged to learn structure-consistent and geometry-aware representations, thereby improving robustness to appearance variations across different sensors and spatial resolutions. Conventional CLIP image encoders based on ViT typically generate fixed-size, low-resolution feature maps, which limits their ability to capture fine-grained spatial structures in remote sensing images. To address this limitation, we adopt the Swin Transformer [38] as the backbone of RSIE, which provides hierarchical and multiscale feature representations while maintaining computational efficiency.

As shown in Fig. 2(b), given a set of batch (B) input image $I \in \mathbb{R}^{B\times H\times W\times 3}$, we first divide it into nonoverlapping 4× four patches through a patch embedding layer and linearly project each patch into a 48-D feature vector, yielding the initial patch embeddings $P^I \in \mathbb{R}^{B\times 48\times H/4\times W/4}$. Following SimMIM [39], we apply a random masking strategy to P^I. Instead of sampling masks directly at the original resolution, masking is first generated at the coarsest spatial scale. Specifically, we randomly generate a binary mask $M_{low} \in \{0,1\}^{B\times H/16\times W/16}$ at resolution $H/16 \times W/16$ with a fixed masking ratio r. Same as the original SimMIM setting, r is set to 0.6. Then, we upsample M_{low} to match the resolution of P^I via nearest-neighbor replication:

$$M = Repeat\,(M_{low}, 4, 4) \in \{0,1\}^{B\times \frac{H}{4}\times \frac{W}{4}} \tag{1}$$

So that each mask unit corresponds to a 4×4 patch block. The masked patch features are replaced by a learnable mask embedding e_{mask}, yielding

$$P^I_{mask} = (1-M)\odot P^I + M\odot e_{mask} \tag{2}$$

where $\odot$ denotes element-wise multiplication with channel-wise broadcasting. This masking strategy encourages the model to utilize broader contextual information to reconstruct occluded regions.

To ensure that the final output features are spatially aligned with CLIP-based features, we employ three standard Swin Transformer stages for contextual modeling of P^I_{mask}. The output feature sizes of each stage are $F^1_{rs} \in \mathbb{R}^{B\times C_1\times H/4\times W/4}$, $F^2_{rs} \in \mathbb{R}^{B\times C_2\times H/8\times W/8}$, and $F^3_{rs} \in \mathbb{R}^{B\times C\times H/16\times W/16}$, where F^3_{rs} serves as the final encoded feature for reconstruction.

During pretraining, we introduce a lightweight reconstruction decoder to restore the masked pixel values. The decoder first applies a 1×1 convolution to project the feature dimension from C to $768 = 16\times 16\times 3$. Then, a pixel shuffle operation with scale factor 16 is used to upsample the feature map, reconstructing the three-channel RGB image $I_{mask} \in \mathbb{R}^{B\times H\times W\times 3}$.

The reconstruction loss $\mathcal{L}_{mask}$ is defined as the L1 loss between the reconstructed image I_{mask} and the original image I over masked regions:

$$\mathcal{L}_{mask} = \frac{1}{|\Omega_{mask}|}\sum_{(x,y)\in\Omega_{mask}} \|I_{mask}(x,y) - I(x,y)\|_1 \tag{3}$$

where $\Omega_{mask} = \{(x,y)\,|\,M_{high}(x,y)=1\}$ denotes the set of masked pixels, and M_{high} is obtained by extending M through nearest neighbor assignment:

$$M_{high} = Repeat\,(M,4,4) \in \{0,1\}^{B\times H\times W} \tag{4}$$

Notably, the reconstruction decoder is used only during pretraining and is discarded in all downstream tasks, ensuring that RSIE remains lightweight and efficient during inference.

C. Cost Map Construction

CLIP aligns visual and text representations in a shared embedding space via contrastive learning, enabling explicit measurement of semantic similarity. We utilized this property to construct a cost map that links visual perception with open vocabulary semantics.

We compute cosine similarity between $F^3_{clip} \in \mathbb{R}^{B\times C\times H/16\times W/16}$ and $F_{class} \in \mathbb{R}^{B\times N\times P\times C}$:

$$S(x,y,n) = \frac{F_{class}(n)\cdot F^3_{clip}(x,y)}{\|F_{class}(n)\|\left\|F^3_{clip}(x,y)\right\|} \tag{5}$$

The $S(x,y,n) \in \mathbb{R}^{B\times N\times P\times H/16\times W/16}$ representing semantic similarity between each spatial location (x,y) and each class n.

In open vocabulary settings, the N varies across training and inference. To enable a class-agnostic implementation and facilitate prompt-level feature aggregation, we merge the B and N and reshape the similarity as $\tilde{S}(x,y,n) \in \mathbb{R}^{(B\times N)\times P\times H/16\times W/16}$. The prompt dimension ($P$) is then treated as the input channel dimension of a 7×7 convolution. The convolution aggregates local spatial context while projecting the prompt-level similarity responses into a compact (C')-dimensional representation, producing the cost map $F_{cost} \in \mathbb{R}^{(B\times N)\times C'\times H/16\times W/16}$:

$$F_{cost} = Conv2d_{7\times 7}\,(Reshape\,(S\,(x,y,n))) \tag{6}$$

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D. Semantic-Structural Cost Map Enhancer

Although F_{cost} captures class-level semantic correlations, it is still susceptible to noise caused by scene heterogeneity and local ambiguity. This limitation primarily arises from the inherent characteristics of CLIP features, which emphasize global semantics but lack sufficient sensitivity to the fine-grained spatial structures that are crucial for remote sensing images.

To address this issue, we propose a SSCME to progressively refine F_{cost}, thereby improving semantic consistency and spatial structural perception. As shown in Fig. 2(a), a complete enhancement cycle consists of three sequential components: global semantic consistency enhancement (GSCE), local spatial structure enhancement (LSSE), and class response enhancement (CRE). SSCME performs this complete enhancement process m times, yielding the refined cost map $F_{cost}^{m} \in \mathbb{R}^{(B\times N)\times C'\times H/16\times W/16}$. The three components are described in detail below.

1) Global Semantic Consistency Enhancement: It utilizes F_{clip}^{3} to reinforce the global semantic consistency of F_{cost}. This module emphasizes reliable global semantic priors, which are beneficial for suppressing scene-specific noise. First, the channel dimension of F_{clip}^{3} is projected via a 1×1 convolution to match that of F_{cost}, and then broadcast along the class dimension and reshape to obtain $\tilde{F}_{clip}^{3} \in \mathbb{R}^{(B\times N)\times C'\times H/16\times W/16}$:

$$\tilde{F}_{clip}^{3} = Reshape\left(Broadcast_{N}\left(Conv2d_{1\times 1}\left(F_{clip}^{3}\right)\right)\right) \tag{7}$$

As shown in Fig. 4(a), the $\tilde{F}_{clip}^{3}$ is normalized using layer normalization (LN) and employed as a guiding feature for enhancing F_{cost} through a Swin Transformer block:

$$F'_{cost} = Swin\left(F_{cost}, LN\left(\tilde{F}_{clip}^{3}\right)\right) \tag{8}$$

Specifically, within the Swin Transformer block, the latest cost map and the guiding feature are concatenated before both the window-based multihead self-attention (W-MSA) and the shifted W-MSA (SW-MSA), enabling joint modeling of global semantic priors and pixel-level cost representations.

2) Local Spatial Structure Enhancement: It further enhances local spatial structures by modeling spatial dependencies using F_{rs}^{3}. Similar to GSCE, the channel dimension of F_{rs}^{3} is first projected via a 1×1 convolution to match that of F'_{cost} and then broadcast along the class dimension and reshape to obtain $\tilde{F}_{rs}^{3} \in \mathbb{R}^{(B\times N)\times C'\times H/16\times W/16}$:

$$\tilde{F}_{rs}^{3} = Reshape\left(Broadcast_{N}\left(Conv2d_{1\times 1}\left(F_{rs}^{3}\right)\right)\right) \tag{9}$$

The broadcast structural feature is then normalized and used as guidance in another Swin Transformer block:

$$F''_{cost} = Swin\left(F'_{cost}, LN\left(\tilde{F}_{rs}^{3}\right)\right) \tag{10}$$

Similarly, the latest cost map and guiding structural feature are concatenated before W-MSA and SW-MSA within the Swin block. It is worth noting that although GSCE and LSSE adopt the same Swin Transformer enhancement architecture, LSSE operates on the semantically refined cost map produced by GSCE and employs a different guiding feature.

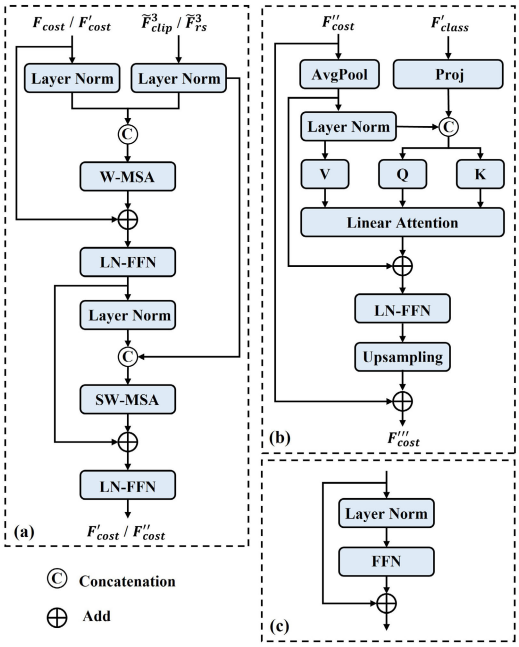


Fig. 4. (a) Architecture of GSCE and LSSE, (b) architecture of CRE, and (c) LN-FFN.

3) Class Response Enhancement: To further strengthen interclass discrimination, CRE explicitly models semantic relationships among classes using class embedding features F_{class}.

First, class embeddings from different text templates are aggregated via mean pooling to obtain $F'_{class} \in \mathbb{R}^{B\times N\times C}$. A linear projection and reshape operation is then applied to align the embedding dimension with that of F''_{cost} to obtain $\tilde{F}'_{class} \in \mathbb{R}^{(B\times N)\times C'}$:

$$\tilde{F}'_{class} = Reshape\left(Proj\left(F'_{class}\right)\right) \tag{11}$$

As shown in Fig. 4(b), a linear transformer block is adopted for class-aware enhancement:

$$F'''_{cost} = Linear\left(V, Concat\left(V, \tilde{F}'_{class}\right)\right) \tag{12}$$

where the concatenation of V and $\tilde{F}'_{class}$ as key matrix and query matrix, while V as value matrix. The V is obtained via spatial mean pooling followed by LN:

$$V = LN\left(AvgPool\left(F''_{cost}\right)\right) \tag{13}$$

E. Dual-Prior Guided Decoder

To effectively convert the refined cost map F_{cost}^{m} into high-resolution segmentation results, we design a DPGD. In the process of gradually restoring spatial resolution, the decoder adaptively integrates semantic priors from CLIP and structural priors from RSIE, thereby achieving a synergistic improvement of semantic consistency and spatial refinement. As shown in Fig. 2(a), at each decoding stage, an efficient spatial gate fusion (ESGF) module is introduced. ESGF serves as a reusable stage-level module that dynamically weights CLIP and RSIE features at different spatial resolutions, providing spatially aware guidance for subsequent cost map decoding. As shown in Fig. 5, ESGF learns spatially varying fusion weights,

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achieving a dynamic balance between semantic alignment capability and preservation of spatial details.

At the ith decoding stage ($i \in \{2, 1\}$), ESGF first upsamples F^i_{clip} to match the spatial resolution of the corresponding F^i_{rs}. Both features are then processed by 33 convolutions for feature extraction and channel compression:

$$\tilde{F}^i_{clip} = Conv2d_{3\times3}\left(up\left(F^i_{clip}\right)\right) \tag{14}$$

$$\tilde{F}^i_{rs} = Conv2d_{3\times3}\left(F^i_{rs}\right) \tag{15}$$

where $\tilde{F}^i_{clip}, \tilde{F}^i_{rs} \in \mathbb{R}^{B\times C^i_d\times H/(4\times i)\times W/(4\times i)}$.

The two features are concatenated along the channel dimension and processed by a depthwise separable convolution ($DWConv$) to model feature interactions:

$$F^i_{cat} = DWConv_{3\times3}\left(Concat\left(\tilde{F}^i_{clip}, \tilde{F}^i_{rs}\right)\right) \tag{16}$$

Then, ESGF predicts adaptive spatial weights for CLIP and RSIE features:

$$w^i = Softmax\left(Conv2d_{1\times1}\left(F^i_{cat}\right)\right) \tag{17}$$

where $w^i \in \mathbb{R}^{B\times2\times H/(4\times i)\times W/(4\times i)}$, corresponding to semantic and structural prior weights, respectively.

The spatially fused guidance feature is finally computed as:

$$F^i_{sf} = w^i[:,0] \odot \tilde{F}^i_{clip} + w^i[:,1] \odot \tilde{F}^i_{rs} \tag{18}$$

To fuse with the enhanced cost map, $F^i_{sf} \in \mathbb{R}^{B\times C^i_d\times H/(4\times i)\times W/(4\times i)}$ is broadcasted along the class dimension and reshaped to obtain $\tilde{F}^i_{sf} \in \mathbb{R}^{(B\times N)\times C^i_d\times H/(4\times i)\times W/(4\times i)}$:

$$\tilde{F}^i_{sf} = Reshape\left(Broadcast_N\left(F^i_{sf}\right)\right) \tag{19}$$

Next, at the ith decoding stage, transposed convolution is applied to upsample and compress channel dimensions:

$$F^i_{up} = \begin{cases} ConvTranspose2d\left(F^m_{cost}\right), & i = 2 \\ ConvTranspose2d\left(F^{i+1}\right), & i = 1 \end{cases} \tag{20}$$

The F^i_{up} shares the same spatial resolution as $\tilde{F}^i_{sf}$. They are concatenated along the channel dimension and fed into a convolution-based multilayer perceptron (MLP) for feature refinement:

$$F^i = MLP\left(Concat\left(F^i_{up}, F^i_{sf}\right)\right) \tag{21}$$

The resulting feature $F^i \in \mathbb{R}^{(B\times N)\times\tilde{C}^i_d\times H/(4\times i)\times W/(4\times i)}$ is passed to the next decoding stage, progressively restoring spatial resolution and refining object boundaries.

After the final decoding stage, a 3×3 convolution is applied to project the channel dimension $\tilde{C}^i_d$ to a single class-response score for each class-specific feature map:

$$F_{logit} = Conv2d_{3\times3}\left(F^1\right) \tag{22}$$

where $F_{logit} \in \mathbb{R}^{(B\times N)\times1\times H/4\times W/4}$. The singleton channel dimension is then removed and the tensor is reshaped to recover the batch and class dimensions to obtain $F'_{logit} \in \mathbb{R}^{B\times N\times H/4\times W/4}$. Finally, the low-resolution logits are upsampled to the original image resolution to obtain $F'_{logit} \in \mathbb{R}^{B\times N\times H\times W}$.

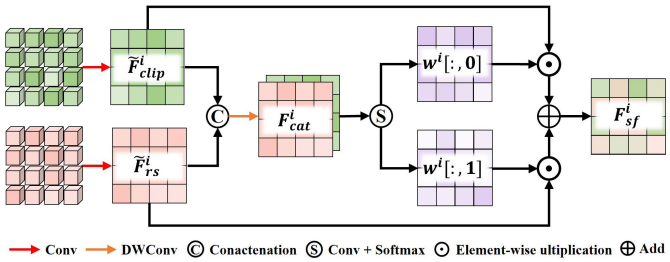


Fig. 5. Architecture of the ESGF module.

IV. Experimental Settings

A. Datasets

Table II summarizes the key characteristics of the datasets used for training and inference. These classes sharing identical semantic definitions with the training set are regarded as seen classes, whereas classes with different semantic definitions are considered unseen classes. The background class is reported separately as a partially matched category because it encompasses diverse land-cover types whose semantic composition varies across datasets. The proposed RS-OVSGSeg model is trained on the OpenEarthMap dataset, which provides high-quality and detailed pixel-wise annotations. To rigorously evaluate the open-vocabulary scene generalization capability of the proposed method, we conduct inference on five additional datasets that are completely unseen during training.

1) Potsdam dataset (https://www.isprs.org/resources/datasets/benchmarks/UrbanSemLab/Default.aspx) is a typical European historical city with large building complexes, narrow streets, and dense settlement structures. Its ultrahigh spatial resolution poses a significant challenge for models trained on satellite images. The inference set consists of 14 images, each with a spatial size of 6000×6000 pixels.
2) LoveDA dataset [40] is a land-cover domain-adaptive semantic segmentation dataset containing high-resolution images from both urban and rural areas in Nanjing, Guangzhou, and Wuhan. The inference set includes 1669 images, each of size 1024×1024 pixels.
3) BLU dataset [41] was collected in June 2018 over Beijing by the Beijing-2 satellite, provided by the 21st Century Aerospace Technology Company Ltd. The scenes cover both urban and rural environments. The inference set comprises 60 images of size 2048×2048 pixels.
4) IRSAMap dataset [42] covers 79 representative regions across six continents, with a total area of approximately 1000 km^2. It spans nine first-level ecological zones in China as well as multiple global regions, ensuring strong diversity and representativeness. The inference set contains 1252 images of size 1024×1024 pixels.
5) FLAIR #1 dataset [43] is part of the dataset currently used at IGN to establish the French national land cover map reference. It contains 50 different spatial domains with high spatiotemporal heterogeneity. The inference set consists of 15 700 images of size 512×512 pixels.

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TABLE II
SUMMARY OF THE TRAINING AND EVALUATION DATASETS

Dataset	Year	Train	Inference	Spatial resolution	Source	Seen classes	Unseen Classes
OpenEarthMap	2023	✓		0.25-0.5m	Satellites and aerial	bareland, rangeland, developed land, building, background	space, road, tree, water, agriculture
Potsdam	2013		✓	0.05m	Aerial	building, tree	impervious surface, low vegetation, car
LoveDA	2021		✓	0.3 m	Google Earth	building, road, water, agricultural, barren,	forest
BLU	2022		✓	0.8 m	Beijing-2	water, road, agricultural,	built up, vegetation
IRSAMap	2025		✓	0.5m	Jilin-1, Google Earth	tree, building, road	farmland, grass, river, lakes, sea, sport
FLAIR #1	2022		✓	0.2 m	Ultracam Eagle M3 and CAMv 2	building, water, agricultural land, bare soil	pervious surface, impervious surface, coniferous, deciduous, brushwood, vineyard, herbaceous vegetation, plowed land

Note: The background class in each inference dataset include complex and diverse land cover types, so it is not classified as either a seen or unseen class but is instead treated as a separate partially matched class for statistical analysis.

We further performed an image overlap analysis between the pretraining and evaluation datasets to assess the possibility of potential data leakage. First, perceptual hashes (pHash) were computed for all images in USGMS-100K and the downstream evaluation datasets. Candidate image pairs with Hamming distances below a threshold of 2 were manually inspected, and no duplicated or near-duplicated image pairs were identified. Second, for datasets with publicly available geographic information, we further verified that their geographic regions do not overlap with those of the downstream evaluation datasets. However, since several public datasets do not provide geolocation metadata, complete geographic independence between the pretraining and evaluation datasets cannot be fully verified. These analyses substantially reduce the likelihood of data leakage between the pretraining and evaluation datasets.

B. Implementation Details

RSIE was pretrained on a Linux PC with four NVIDIA GeForce RTX A6000 GPUs. We used a batch size of 128 and trained the model for 300 epochs. To enhance robustness across spatial resolutions, we applied random cropping and random horizontal flipping as data augmentation strategies. We employed the AdamW optimizer with a cosine learning rate scheduler, an initial learning rate of 5×10^{-5}, weight decay of 0.05, and a 30 epoch linear warm-up phase.

The experiment of RS-OVSGSeg was conducted on a Linux PC with four NVIDIA GeForce RTX 4090D GPUs. During the training process, we used random flipping for data augmentation. Using the AdamW optimizer with polynomial decay schedule, setting the initial learning rate to 2×10^{-5}, weight decay to 1×10^{-4}, and momentum parameters to $\beta = (0.9, 0.999)$. The model is trained using cross-entropy loss. The CLIP image encoder was based on the ViT-B/16 architecture. The batch size was set to 4, and the maximum number of training iterations was 80 000. During inference, we applied a sliding-window strategy with a window size of 384×384 pixels and a stride of 288×288 pixels to mitigate boundary artifacts and ensure complete spatial coverage.

C. Evaluation Metrics

We report mean $F1$-score ($mF1$) and mean Intersection over Union ($mIoU$) for each dataset, which are the most widely used metrics in RSISS. These two metrics comprehensively reflect the performance of the model from different dimensions. In addition, we also calculated the seen class mean IoU ($smIoU$), unseen class mean IoU ($umIoU$), and their harmonic mean $hIoU$, *mean* $mF1$ and *mean* $mIoU$ to evaluate the average performance of the model on multiple datasets. The metrics are defined as follows:

$$mF1 = \frac{1}{N}\sum_{k=1}^{N}\frac{2 \times Precision_k \times Recall_k}{Precision_k + Recal_k} \tag{23}$$

$$Precision_k = \frac{TP_k}{FP_k + TP_k} \tag{24}$$

$$Recall_k = \frac{TP_k}{FN_k + TP_k} \tag{25}$$

$$mIoU = \frac{1}{N}\sum_{k=1}^{N} IoU_k = \frac{1}{N}\sum_{k=1}^{N}\frac{TP_k}{TP_k + FP_k + FN_k} \tag{26}$$

$$smIoU = \frac{1}{|C_s|}\sum_{k\in C_s} IoU_k \tag{27}$$

$$umIoU = \frac{1}{|C_u|}\sum_{k\in C_u} IoU_k \tag{28}$$

$$hmIoU = \frac{2 \times smIoU \times umIoU}{smIoU + umIoU} \tag{29}$$

where TP_k, FP_k, TN_k, and FN_k denote true positives, false positives, true negatives, and false negatives for class k, respectively. N is the number of classes in a dataset, C_s, C_u = the sets of seen and unseen classes.

V. EXPERIMRNTAL RESULTS

A. Comparison With State-of-the-Art Methods

To comprehensively evaluate the performance of the proposed RS-OVSGSeg, we designed a series of comparative experiments. The compared methods are all the latest OVSS methods in the field of remote sensing, including OVRS [21] and Skysense-O [10], which are based on CLIP-trained architectures, TPOV-Seg [19], which incorporates GPT-assisted synonym expansion for class representation, SegEarth-OV3 [16], a training-free method based on SAM3, GSNet [11] and RSKT-Seg [12], which introduce a DINO-based RSIE backbone, and AeroSeg [22], which integrates SAM-guided features. For fairness, all compared methods are implemented

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TABLE III

QUANTITATIVE COMPARISON OF DIFFERENT METHODS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Method	$mF1/mIoU$ (%)					*mean mF1* (%)	*mean mIoU* (%)
	Potsdam	LoveDA	BLU	IRSAMap	FLAIR #1		
Cat-Seg	56.57/44.58	50.79/36.82	46.99/33.09	47.78/36.09	35.85/25.68	47.60	35.25
OVRS	55.83/43.92	50.89/36.96	55.86/32.69	52.36/35.38	39.07/28.16	50.80	35.42
TPOV-Seg	**59.55**/**46.57**	52.5/37.94	41.46/29.07	46.96/35.31	41.63/30.88	48.42	35.96
SegEarth-OV3	49.14/37.51	**56.84**/40.20	38.30/25.28	41.90/26.62	22.52/15.30	41.74	28.98
Skysense-O	45.27/35.35	55.63/**41.84**	53.21/**39.65**	48.07/36.89	33.52/25.20	47.14	35.79
GSNet	56.85/45.02	51.04/37.21	**56.95**/33.50	47.94/36.74	35.89/26.27	49.73	35.75
AeroSeg	53.77/41.27	50.78/36.92	50.04/36.13	44.98/33.33	36.86/26.24	47.28	34.78
RSKT-Seg	57.87/46.49	52.06/38.06	46.62/32.73	45.63/34.38	35.60/25.86	47.56	35.50
Ours	59.13/45.65	51.67/37.32	52.41/37.07	**54.62**/**40.34**	**44.25**/**32.73**	**52.42**	**38.62**

TABLE IV

PERFORMANCE COMPARISON ON SEEN, UNSEEN, AND PARTIALLY MATCHED CATEGORIES ACROSS DIFFERENT INFERENCE DATASETS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Method	$smIoU/umIoU/partially\ matched\ IoU$ (%)					*mean smIoU* (%)	*mean umIoU* (%)	*mean hmIoU* (%)
	Potsdam	LoveDA	BLU	IRSAMap	FLAIR #1			
Cat-Seg	67.82/43.94/0.01	45.65/22.61/6.92	36.17/45.02/0.00	58.88/30.71/0.00	47.25/18.10/0.03	51.15	32.07	39.43
OVRS	67.64/42.70/0.18	45.65/23.53/6.92	37.38/42.00/0.00	58.64/29.65/0.00	48.86/21.17/1.20	51.63	31.81	39.37
TPOV-Seg	61.83/**48.63**/9.88	44.37/30.76/12.97	21.54/46.29/**17.22**	59.60/29.05/0.00	**53.19**/23.49/0.82	48.11	35.64	40.95
SegEarth-OV3	49.41/40.48/4.82	40.77/32.41/**45.16**	31.86/20.31/15.50	52.20/15.44/16.99	29.57/10.01/0.47	40.76	23.73	30.00
Skysense-O	65.64/26.32/1.87	**50.08**/**35.59**/6.93	**49.18**/44.75/0.88	60.16/31.21/1.14	44.58/18.53/1.02	**53.93**	31.28	39.59
GSNet	64.99/46.61/0.29	45.62/25.44/6.92	37.70/43.96/0.00	**61.59**/30.44/0.00	46.68/19.17/1.38	51.32	33.12	40.26
AeroSeg	66.28/37.91/1.30	46.72/17.92/6.95	39.62/48.88/0.15	56.82/27.14/0.03	48.61/17.80/**4.27**	51.61	29.93	37.89
RSKT-Seg	**68.70**/47.17/0.03	47.03/24.38/6.93	37.87/41.38/0.00	59.49/27.55/0.00	46.17/18.70/1.90	51.85	31.83	39.45
Ours	64.09/44.42/**12.43**	45.12/24.72/10.89	36.26/**49.59**/14.14	58.97/**33.60**/**24.94**	52.28/**26.94**/0.78	51.35	**35.85**	**42.22**

following their publicly released configurations and official settings, including backbone architectures, pretrained weights, and prompt templates.

Table III presents the quantitative evaluation results of different methods. It can be observed that our method achieves 52.42% *mean mF*1 and 38.62% *mean mIoU*, obtaining the best overall performance among all compared methods. TPOV-Seg achieves the second-best performance by applying GPT-based synonym expansion to enrich class semantics in advance. Under the same class setting, Skysense-O outperforms other methods that introduce additional remote sensing feature extractors (e.g., GSNet, AeroSeg, and RSKT-Seg), mainly due to its use of large-scale architectures such as ViT-L and Swin-H. In contrast, SegEarth-OV3 relies on the publicly available SAM3 model, which exhibits the lowest average performance among all methods. The outstanding performance on these datasets validates the strong scene generalization capability of RS-OVSGSeg.

Furthermore, Table IV reports the *mIoU* of the seen, unseen, and partially matched class groups, together with the *mean smIoU*, *mean umIoU*, and *mean hmIoU*. As shown in Table IV, our method achieves the highest *mean umIoU* of 35.85% while maintaining a competitive *mean smIoU* of 51.35%. More importantly, our method obtains the best *mean hmIoU* of 42.22%, outperforming the second-best method by 1.27%. These results demonstrate that the proposed method achieves a better balance between recognizing previously seen classes and generalizing to unseen classes, rather than improving unseen-class performance at the expense of seen classes. The superiority of our method is particularly evident on IRSAMap and FLAIR #1, which contain a large proportion of unseen classes. In contrast, Skysense-O achieves the best performance only on LoveDA, where the number of unseen classes is relatively small and their semantic complexity is lower, while TPOV-Seg performs best only on the Potsdam dataset. For the partially matched background class, our method consistently outperforms all other supervised training methods. This indicates that the proposed semantic-structural collaborative representation is more effective at modeling complex regions containing heterogeneous land-cover types and ambiguous semantic boundaries, thereby reducing the confusion between foreground classes and background regions under large scene variations.

To further analyze the performance of different methods on specific classes, we plotted bar charts of per-class *IoU* across different datasets, as shown in Fig. 6. It can be observed that almost all methods exhibit similar performance on relatively simple datasets with fewer classes, such as Potsdam, LoveDA, and BLU. Moreover, their prediction accuracy in the classes of building, road, and agricultural are basically at the same level. However, on more complex datasets with finer-grained classes, such as IRSAMap and FLAIR #1, different methods show considerable variability in segmentation performance across classes. Our proposed RS-OVSGSeg method consistently achieves competitive or superior performance on most unseen classes across different datasets. For example, our method obtains the highest IoU on unseen classes such as sport in IRSAMap and vineyard, herbaceous vegetation, and plowed land in FLAIR #1. These classes exhibit substantial appearance variations and limited semantic overlap with the training

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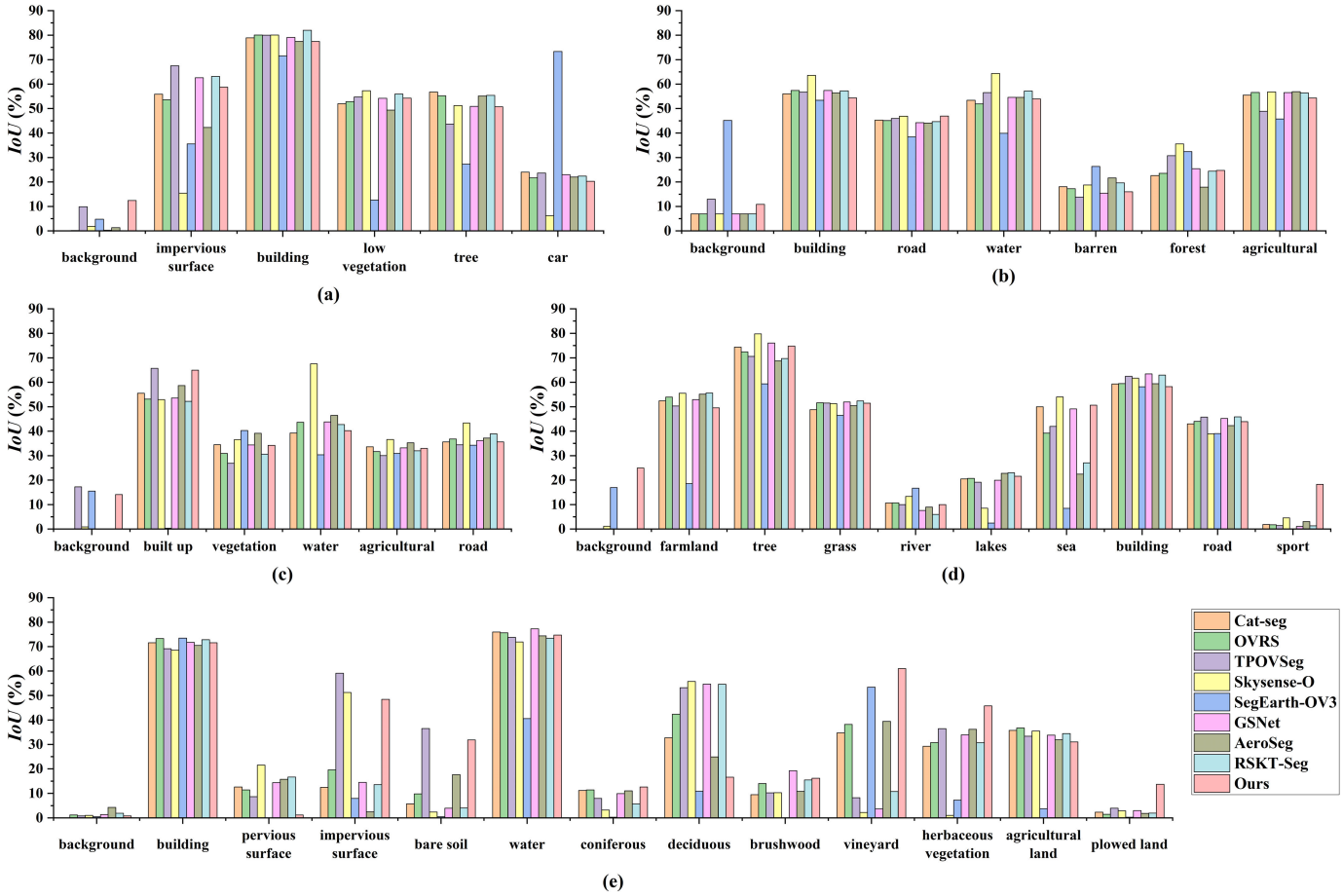


Fig. 6. Per-class *IoU* scores of different methods across datasets. (a) Potsdam dataset, (b) LoveDA dataset, (c) BLU dataset, (d) IRSAMap, and (e) FLAIR #1 dataset.

classes, making them particularly challenging under the open-vocabulary setting. The superior performance on these unseen classes demonstrates that the proposed semantic–structural collaborative framework effectively transfers semantic knowledge to novel classes while preserving discriminative structural representations under large scene variations. It is worth noting that SegEarth-OV3 achieves remarkably high accuracy on the car in the Potsdam dataset, surpassing other methods by a large margin. This indicates that SAM3 exhibits strong capability in segmenting small objects commonly appearing in natural images. However, its performance is extremely poor on remote sensing-specific classes with significant scale variations, such as low vegetation, impervious surface, pervious surface, and agricultural land.

Fig. 7 visualizes segmentation results of different methods on multiple cross-scene datasets. Our method generates more coherent and visually consistent segmentation maps, with clearer object boundaries and significantly fewer noisy or fragmented predictions. In contrast, baseline methods often suffer from semantic confusion and spatial inconsistency, particularly in regions with complex textures or small objects. Specifically, methods relying solely on CLIP architecture tend to produce coarse and over-smoothed predictions, frequently confusing different land cover classes with similar semantic descriptions. This phenomenon indicates that global semantic alignment alone is insufficient for fine-grained dense prediction in remote sensing images. Methods incorporating external domain features exhibit better semantic discrimination in certain regions. However, their predictions still suffer from blurred boundaries, especially around object contours and small-scale structures such as narrow roads, scattered buildings, and fragmented vegetation patches. This limitation is mainly caused by the fixed-resolution multiscale features of ViT-style encoders, which struggle to capture detailed multiscale spatial structures. In contrast, our method demonstrates superior boundary preservation and spatial continuity. Benefiting from the hierarchical multiscale features provided by the RSIE and the semantic–structural cost map enhancement mechanism, our model accurately delineates object contours while maintaining consistent semantic responses across large and complex scenes.

B. Ablation Study

To evaluate the effectiveness of different modules in the proposed framework, all ablation experiments are conducted under identical settings, including the same CLIP backbone, training dataset, prompt templates, inference strategy, and hardware environment. We use an architecture that uses CLIP and RSIE features for decoding and does not use SSCME

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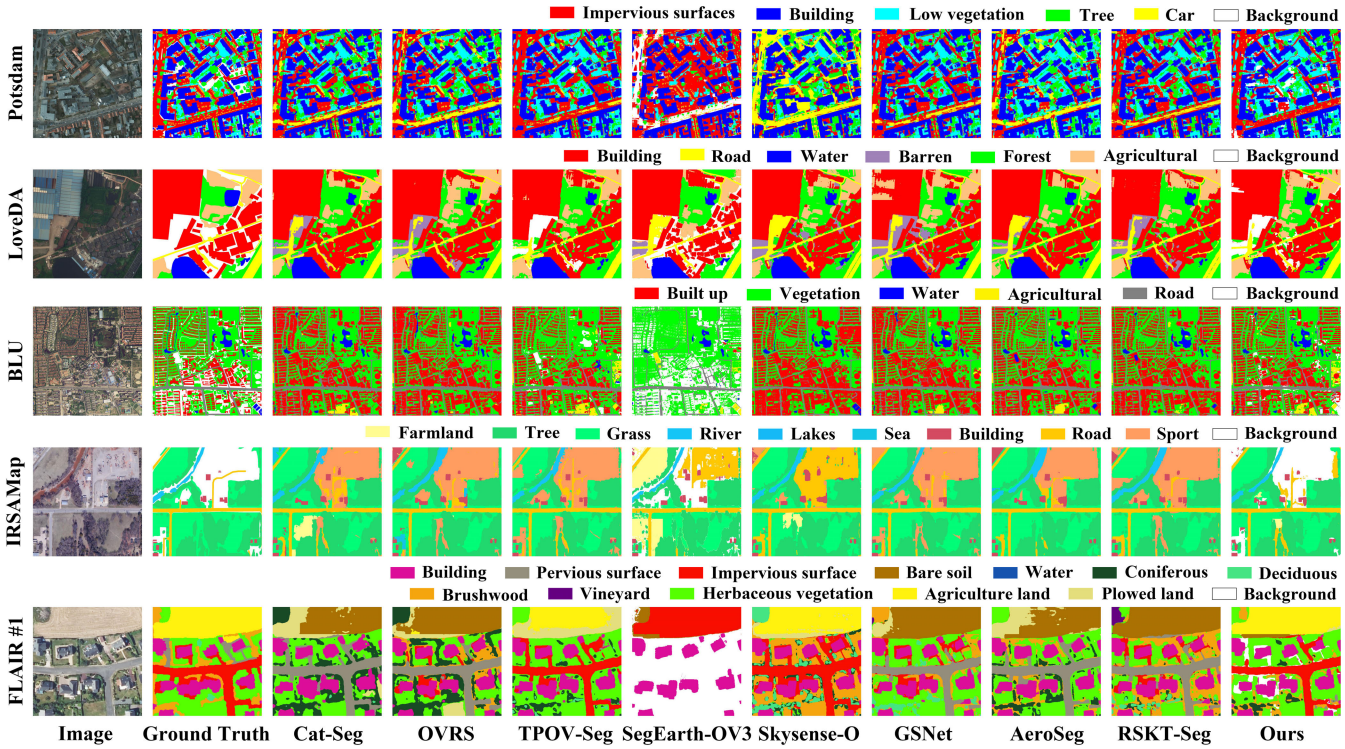


Fig. 7. Qualitative comparison of segmentation results across datasets.

TABLE V
RESULTS OF ABLATION STUDIES ON DIFFERENT MODULES

SSCME			DPGD	$mF1/mIoU$ (%)					*mean mF*1	*mean mIoU*
GSCE	LSSE	CRE		Potsdam	LoveDA	BLU	IRSAMap	FLAIR #1	(%)	(%)
				54.95/41.29	49.03/35.31	46.62/32.93	44.73/32.14	42.51/30.60	47.57	34.45
✓				57.03/42.96	49.88/36.02	49.27/34.96	46.66/33.77	42.69/30.97	49.11	35.73
✓	✓			56.00/42.53	50.45/36.26	50.88/35.73	48.00/34.94	46.00/33.73	50.27	36.64
✓	✓	✓		55.46/42.57	51.24/36.97	52.68/37.64	49.89/35.99	46.61/34.67	51.19	37.57
✓	✓	✓	✓	59.13/45.65	51.67/37.32	52.41/37.07	54.62/40.34	44.25/32.73	52.42	38.62

and DPGD modules as the baseline model, gradually adding GSCE, LSSE, CRE, and DPGD modules.

Table V summarizes the quantitative results under different ablation configurations. It can be observed that the introduction of each module can further improve segmentation performance. Specifically, the baseline model achieves 47.57% *mean mF*1 and 34.45% *mean mIoU*. After applying the SSCME module for cost map enhancement, *mean mF*1 and *mean mIoU* increase by 3.62% and 3.12%, respectively. Among its submodules, GSCE contributes the largest improvement (+1.54% *mean mF*1, +1.28% *mean mIoU*), followed by LSSE, while CRE has the smallest improvement. This indicates that cost maps constructed solely from CLIP alignment features are too coarse for direct decoding. By enhancing global semantics, local structures, and class features, noise semantics can be effectively suppressed and spatial consistency can be enhanced, achieving reliable dense prediction in complex remote sensing scenes. When DPGD is used to adaptively balance semantic priors from CLIP and structural priors from RSIE, the model performance improved by 1.23% and 1.05%, respectively. This indicates that guiding features at different scales can to some extent compensate for complementary information that CLIP cannot capture. Meanwhile, effective feature adaptive fusion methods can greatly improve the segmentation performance of the model.

We visualized the intermediate features of the model in a heatmap under different configurations, as shown in Fig. 8. After introducing RSIE features for decoding, background and foreground activations become more distinguishable, and erroneous activations in building and road classes are suppressed. After introducing GSCE for global consistency enhancement, it can be seen that responses of herbaceous vegetation and impervious surface become more spatially coherent, significantly reducing scattered noise caused by complex scene structures. LSSE further corrected some erroneous areas by combining local RSIE features, enhanced background response suppression, and formed clearer object boundaries. By introducing CRE, foreground activations are notably strengthened, particularly for impervious surface and bare soil, indicating that semantic relational constraints help alleviate confusion between visually similar but semantically different classes. Finally, DPGD adaptively learns fusion weights between

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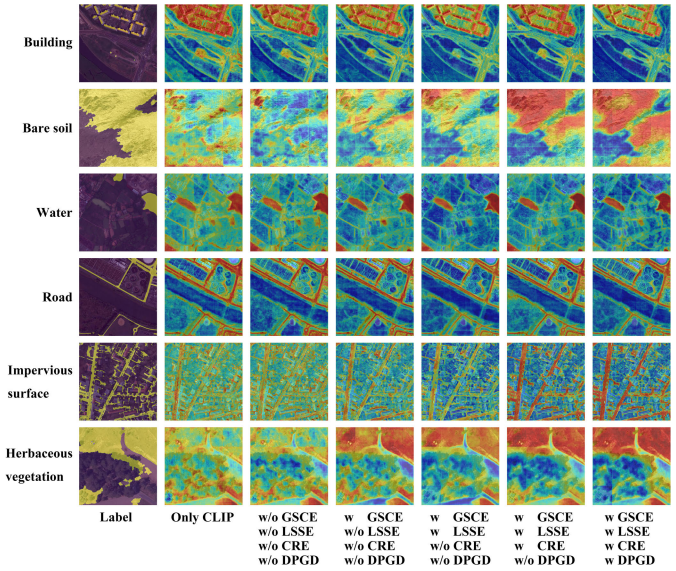


Fig. 8. Feature heatmaps of each module in the proposed method.

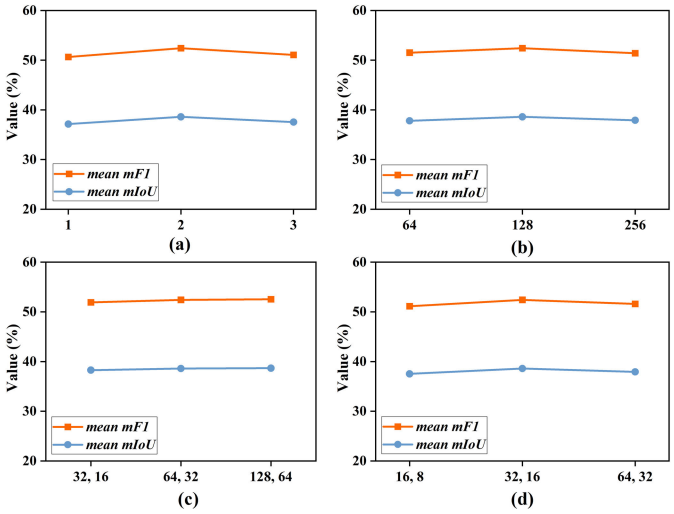


Fig. 9. Performance variations under different hyperparameter settings. (a) Number of cost map enhancement iterations m, (b) channel dimension C' of F_{cost}, (c) feature channel dimension C_d^i of $\tilde{F}_{clip}^i$ and $\tilde{F}_{rs}^i$, and (d) feature channel dimension $\tilde{C}_d^i$ of F^i.

TABLE VI
COMPARISON OF DIFFERENT FEATURE FUSION STRATEGIES

	*mean mF*1 (%)	*mean mIoU* (%)
Add	44.22	31.86
Cat	46.68	33.76
Gated	48.37	35.25
SSCME	52.42	38.62

semantic and structural priors at each spatial location, effectively resolving inconsistencies between global and local features. As a result, the model generates feature maps that preserve both semantic correctness and spatial details. Overall, these modules collaboratively transform the original CLIP-based similarity responses into structurally coherent and semantically reliable segmentation results.

To further validate the advantages of the proposed SSCME, we compare it with three widely used feature fusion strategies under the same experimental settings, including simple addition (Add), feature concatenation (Cat), and conventional gated fusion (Gated). As shown in Table VI, the proposed SSCME achieves the best performance, obtaining 52.42% *mean mF*1 and 38.62% *mean mIoU*. Compared with the strongest conventional gated fusion method among other feature fusion methods, SSCME improves the *mean mF*1 by 4.04% and the *mean mIoU* by 3.37%. The simple addition method has the worst performance. The above results indicate that straightforward feature aggregation is insufficient to effectively exploit the complementary characteristics of semantic and structural representations. The proposed SSCME establishes a more effective interaction between the global semantic features extracted by CLIP and the local structural representations learned by RSIE. Consequently, it produces more discriminative feature representations and leads to superior segmentation performance.

C. Parameter Analysis

Different parameter settings may influence model performance. We analyze four key hyperparameters: number of cost map enhancement iterations m, channel dimension C' of F_{cost}, feature channel dimension C_d^i of $\tilde{F}_{clip}^i$ and $\tilde{F}_{rs}^i$, and feature channel dimension $\tilde{C}_d^i$ of F^i. The results are shown in Fig. 9.

1) Number of Cost Map Enhancement Iterations m: We evaluate performance when $m = 1, 2, 3$, while keeping other parameters fixed. As shown in Fig. 9(a), the best performance is achieved at $m = 2$. When $m = 1$, the enhancement process is not sufficient to fully refine the semantic structure interaction, resulting in decreases of 1.77% in *mean mF*1 and 1.46% in *mean mIoU*. When $m = 3$, excessive refinement resulting in a slight decrease of 1.34% and 1.06% in *mean mF*1 and *mean mIoU*, respectively. This indicates that moderate enhancement depth is more appropriate, and repeated enhancement may introduce noise or excessive smoothing effects.

2) Channel Dimension C' of F_{cost}: A smaller C' limits the representational power of the cost map, while an excessively large C' may introduce redundant features and increase optimization difficulty. We conducted experiments with $C' = 64$, 128, 256. As shown in Fig. 9(b), the model achieves optimal performance when C' is set to 128. Increasing or decreasing the number of channels will slightly decrease the performance of the model by about 0.7%–1%.

3) Feature Channel Dimension C_d^i of $\tilde{F}_{clip}^i$ and $\tilde{F}_{rs}^i$: Fig. 9(c) shows the model performance under three configurations: (32, 16), (64, 32), and (128, 64). It can be observed that this parameter has *a* relatively limited impact on the final performance. The *mean mIoU* only varies within a small range of 38.28% to 38.68%. This indicates that RS-OVSGSeg is insensitive to the channel dimension of intermediate semantic and structural features, reflecting the robustness of the proposed fusion framework.

4) Feature Channel Dimension $\tilde{C}_d^i$ of F^i: As shown in Fig. 9(d), we conducted experiments with $\tilde{C}_d^i$ values of (16, 8), (32, 16), and (64, 32). Best performance is achieved at (32, 16). When $\tilde{C}_d^i = (16, 8)$, *mean mF*1 and *mean mIoU* decrease by 1.77% and 1.08%, respectively, indicating that the small number of channels limits the decoder's expressive capacity. When $\tilde{C}_d^i$ increases to (64, 32), *mean mF*1 and *mean mIoU*

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TABLE VII
MODEL COMPLEXITY AND INFERENCE EFFICIENCY COMPARISON UNDER A FIXED INPUT SIZE OF 384 × 384

Method	Total Params (M)	Train		Inference	
		Trainable Params (M)	FLOPs (G)	GPU memory usage (M)	FPS
Cat-Seg	154.581	25.116	105.875	1418	32.34
OVRS	154.604	25.139	256.917	1420	17.01
TPOV-Seg	154.688	25.222	551.985	1558	29.77
SegEarth-OV3	-	-	-	6108	1.57
Skysense-O	1015.186	1015.186	700.223	4996	13.54
GSNet	245.447	29.007	411.888	2248	15.57
AeroSeg	466.434	28.69	1831.807	5872	4.85
RSKT-Seg	336.502	31.808	581.48	2650	11.17
Ours	215.611	26.723	137.397	1614	15.87

show a slight performance decline and introduce unnecessary complexity.

Overall, the results of the parameter analysis above indicate that RS-OVSGSeg exhibits stable performance under a wide range of hyperparameter settings. The selection of parameters has low sensitivity, and the performance curve remains smooth and performs well under different configurations. This indicates that the proposed framework does not rely on carefully adjusted hyperparameters, but benefits from its inherent semantic structure representation design.

D. Complexity and Efficiency Analysis

In addition to segmentation accuracy, computational complexity and inference efficiency are also critical for practical remote sensing applications. Table VII presents the model parameters (Params), floating-point operations per seconds (FLOPs), GPU memory usage, and frames per second (FPS) for the proposed method and other comparative methods. To ensure a fair comparison, all statistics were measured using a fixed input size of 384 × 384 pixels, with the same class settings, and on an NVIDIA RTX 4090-D GPU.

It can be seen that the proposed method contains 215.611M total parameters, with only 26.723M trainable parameters. Compared with methods relying only on a CLIP encoder, the additional RSIE encoder introduces a moderate increase in parameter count and memory consumption. However, compared with approaches that incorporate large-scale remote sensing backbones, such as GSNet, AeroSeg, and RSKT-Seg, our method maintains significantly lower model complexity while achieving better segmentation performance. Skysense-O adopts a much larger architecture and therefore exhibits the highest parameter count among all compared methods.

In terms of computational complexity, our method requires only 137.397G FLOPs, second only to the Cat-Seg baseline and substantially lower than most compared methods. Although our method does not achieve the highest FPS, the discrepancy between FLOPs and runtime can be explained by the dual-encoder architecture. The additional RSIE encoder introduces extra feature extraction operations, resulting in increased inference latency compared with single-encoder CLIP-based methods. In contrast, the proposed semantic–structural collaboration modules introduce only limited computational overhead. Therefore, the runtime increase mainly originates from the additional pretrained encoder rather than the cost-map refinement process itself. In addition, AeroSeg introduces SAM-guided features and consequently exhibits the highest computational complexity and one of the slowest inference speeds among trainable methods. SegEarth-OV3 relies entirely on the SAM3 foundation model, resulting in significantly higher memory consumption and substantially slower inference.

Overall, the proposed RS-OVSGSeg achieves an effective balance between segmentation performance, computational complexity, and inference efficiency. The introduction of RSIE substantially improves scene generalization capability while maintaining competitive computational cost and practical runtime.

VI. DISCUSSION

A. Performance Analysis of Different Pretrained RSIE Models

To further analyze the contribution of the proposed RSIE pretraining strategy and USGMA-100K dataset, we conducted additional controlled experiments using different pretraining datasets, pretraining scales, backbone architectures, and self-supervised learning paradigms under the same RS-OVSGSeg framework. All experiments were performed using identical training protocols, prompt templates, inference strategies, and hardware settings. The only varying factor was the initialization strategy of the frozen RSIE encoder. The results are summarized in Table VIII.

We first analyze the effect of different pretraining datasets while keeping the Swin-B architecture fixed. Compared with ImageNet pretraining, remote sensing specific pretraining substantially improves scene generalization performance. Specifically, ImageNet-pretrained Swin-B achieves only 47.20% *mean mF*1 and 34.70% *mean mIoU*, whereas pretraining on OpenEarthMap increases the performance to 49.06% and 36.22%, respectively. When the pretraining dataset is further expanded to LandDiscover50K, the performance reaches 51.48% *mean mF*1 and 37.81% *mean mIoU*. Finally, the proposed USGMS-100K pretraining strategy achieves the best performance of 52.42% *mean mF*1 and 38.62% *mean mIoU*. These results indicate that scene-invariant representation learning benefits substantially from large-scale remote sensing pretraining data with diverse geographic regions, sensor types, and imaging conditions. Compared with ImageNet, which mainly contains natural images, remote sensing datasets provide more relevant structural priors. Furthermore, compared with single-source remote sensing datasets, the multisensor composition of USGMS-100K enables RSIE to learn more transferable structural representations under large scene variations.

To further analyze the influence of pretraining dataset scale, we constructed several subsets of USGMS-100K using nested sampling method, containing 20K, 40K, 60K, and 80K image patches, respectively. As shown in Table VIII, the segmentation performance generally improves as the scale of the pretraining dataset increases. Specifically, *mean mIoU* increases from 35.90% using USGMS-20K to 38.62%

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TABLE VIII
COMPARISON OF DIFFERENT RSIE PRETRAINING DATASETS, SCALES, AND SELF-SUPERVISED STRATEGIES

Pre-Training methods	Architecture	Pretrained dataset	*mean mF*1 (%)	*mean mIoU* (%)	Total params (M)	Trainable params (M)	FLOPs (G)
DINO	ViT-B/8	LandDiscover50k	51.76	38.11	245.327	28.887	392.680
MAE	ViT-B/8	USGMS-100K	51.91	38.22	245.327	28.887	294.733
Ours	Swin-B	ImageNet	47.20	34.70			
Ours	Swin-B	OpenEarthMap	49.06	36.22			
Ours	Swin-B	LandDiscover50k	51.48	37.81			
Ours	Swin-B	USGMS-20K	48.44	35.90	215.611	26.723	137.397
Ours	Swin-B	USGMS-40K	48.76	36.10			
Ours	Swin-B	USGMS-60K	49.17	36.45			
Ours	Swin-B	USGMS-80K	49.73	36.80			
Ours	Swin-B	USGMS-100K	52.42	38.62			

TABLE IX
IMPACT OF RSIE ON SEGMENTATION PERFORMANCE

Method	*mean mF*1 (%)	*mean mIoU* (%)
w/o RSIE	50.82	37.26
w RSIE	52.42	38.62

using the complete USGMS-100K dataset, while *mean mF*1 increases from 48.44% to 52.42%. Although the improvements from USGMS-20K to USGMS-80K are relatively moderate, a more noticeable performance gain is achieved when the full USGMS-100K dataset is employed.

We additionally compare different backbone architectures and self-supervised learning strategies. On the LandDiscover50K dataset, the Swin-B-based RSIE is comparable to the DINO-pretrained ViT-B/8 encoder while requiring fewer trainable parameters and substantially lower computational complexity. Similarly, when pretrained on USGMS-100K, the proposed Swin-B-based encoder achieves competitive performance while substantially reducing FLOPs from 294.733G to 137.397G. These results suggest that hierarchical architectures such as Swin Transformer are more suitable for capturing the multiscale structural characteristics than fixed-resolution ViT-based encoders.

Overall, the superior performance of RS-OVSGSeg is not merely due to increased backbone capacity, but rather stems from a representation learning paradigm that is better aligned with the structural characteristics of remote sensing scenes. The combination of USGMS-100K and the Swin-based RSIE enables the learning of scene-invariant structural representations while maintaining favorable computational efficiency.

B. Analysis of the Effectiveness of RSIE

RSIE plays a critical role in the proposed RS-OVSGSeg framework by enhancing structural representation and spatial consistency without interfering with semantic alignment. This design is essential for achieving stable and reliable OVSGSS in remote sensing. In this section, we analyze the role of RSIE from both effectiveness and rationality perspectives.

1) Overall Effectiveness of RSIE: We evaluate the overall effectiveness of RSIE by comparing the full model with a variant that completely removes the RSIE branch. The quantitative

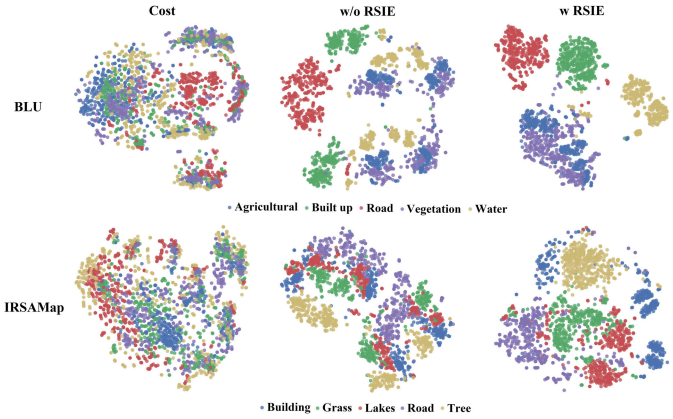


Fig. 10. t-SNE visualization of intermediate feature embeddings with and without RSIE.

results are reported in Table IX. It can be seen that without RSIE, the model achieves 50.82% *mean mF*1 and 37.26% *mean mIoU*. After incorporating RSIE, performance improves to 52.42% and 38.62%, respectively. This consistent gain demonstrates that RSIE provides supplementary information that cannot be captured by CLIP features alone. To further interpret this improvement, we visualize intermediate feature embeddings using t-SNE in Fig. 10. It can be observed that without RSIE, feature distributions of different classes exhibit substantial overlap and loose clustering, indicating weak discriminability in feature space. In contrast, when RSIE is introduced, samples from the same class form more compact clusters, and interclass separation becomes more distinct. These results confirm that CLIP primarily encodes high-level semantic abstraction and global contextual cues but lacks sufficient capacity to model fine-grained spatial structure and geometric layout, which are crucial for dense prediction in remote sensing. RSIE effectively compensates for this limitation by providing structure-aware and geometry-sensitive representations, thereby enhancing feature discriminability and scene generalization.

2) Why RSIE Should Not Participate in Cost Map Computation?: Unlike some existing methods that directly use additional visual encoders for semantic matching, although RSIE brings significant performance improvements, we did

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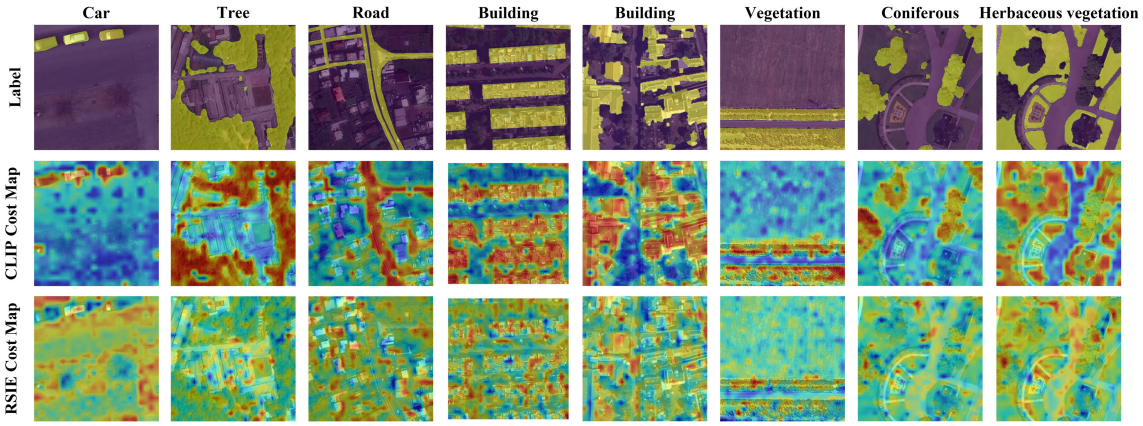


Fig. 11. Comparison of cost maps generated from CLIP features and RSIE features.

TABLE X
EFFECT OF RSIE FEATURES IN COST-MAP COMPUTATION

Method	*mean mF*1 (%)	*mean mIoU* (%)
w/o RSIE cost	52.42	38.62
w RSIE cost	51.44	37.79

not directly use RSIE features for cost map computation. Table X provides quantitative indicators for whether RSIE features are used in cost map calculation. When RSIE features are included, *mean mF*1 drops from 52.42% to 51.44%, and *mean mIoU* decreases from 38.62% to 37.79%. This indicates that directly mixing RSIE features into semantic matching is harmful rather than beneficial. The fundamental reason for this result is the inconsistency in representation between RSIE and CLIP. CLIP features are explicitly aligned with text embeddings through large scale vision–language pretraining, whereas RSIE is trained in a self-supervised manner without language supervision. Therefore, RSIE features and CLIP text features are not in the same semantic embedding space. Fig. 11 further illustrates this phenomenon. Cost maps computed from CLIP features exhibit clear and consistent semantic activation patterns aligned with target classes. In contrast, cost maps derived from RSIE features show noisy and ambiguous responses, failing to provide reliable semantic localization. These observations indicate that although RSIE is highly effective for structural representation learning, it should not be directly used for semantic matching with text embeddings. On the contrary, RSIE is better suited to providing structural priors during feature refinement and decoding process to supplement CLIP's semantic priors without introducing semantic bias.

C. Limitations and Future Work

Despite the proposed framework has strong performance and cross-scene generalization ability, there are still some limitations that deserve further research. First, this study mainly focuses on optical remote sensing images. However, in practical applications, remote sensing interpretation often involves multimodal data sources such as SAR, LiDAR, and multispectral images. Optical images are inherently sensitive to changes in lighting, atmospheric conditions, and seasonal variations, while other modalities provide complementary structural or geometric clues. Extending the proposed OVSGSS framework to a multimodal setting may further enhance robustness, especially in complex regions with ambiguous textures or weak visual contrasts. Secondly, the current RSIE is pretrained in a fixed cross-scene dataset, assuming that target-scene data are entirely unavailable. While this setting evaluates strict scene generalization ability, it does not explore the scenario where unlabeled data from the target region are accessible. A promising direction is to investigate scene-adaptive self-supervised pretraining, where RSIE is first adapted using unlabeled data from the target scene before being integrated into the OVSS framework. This strategy can further reduce domain gaps and improve scene-specific representation capacity. Finally, this work relies on annotated data for training the decoder under an open-vocabulary paradigm. A more challenging yet meaningful direction is to explore fully unsupervised OVSS in remote sensing. Future research may investigate self-training or confidence-guided pseudolabel refinement mechanisms to iteratively improve segmentation performance, rather than relying on labeled datasets. Ultimately, achieving scalable and annotation-free OVSS would significantly broaden the applicability of remote sensing interpretation in large-scale and dynamically evolving environments.

VII. CONCLUSION

In this study, we proposed a unified OVSGSS framework to address the challenges posed by large-scale scene heterogeneity in OVSS tasks. To support robust visual representation learning under substantial cross-scene variations, we construct a large-scale unlabeled dataset, USGMS-100K. And pretrained the Swin Transformer-based RSIE using a mask self-supervised strategy to learn hierarchical, multiscale, and structure aware features. On this basis, we proposed the RS-OVSGSeg framework. The SSCME module are used to refine the semantic matching response, and a DPGD to adaptively integrate semantic and structural representations at multiple scales, effectively overcoming the limitations of existing open-vocabulary methods in modeling complex spatial

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structures and achieving robust scene generalization. Comprehensive experiments on multiple benchmark datasets have shown that RS-OVSGSeg consistently outperforms state-of-the-art methods in both segmentation accuracy and cross-scene generalization. In depth ablation research and qualitative visualization further validated the important role of structural priors and semantic-structure collaboration in OVSS. Overall, this work provides new insights into the semantic–structural representation learning of OVSS and establishes a promising foundation for future research toward more generalizable, scalable, and adaptable remote sensing interpretation systems.

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